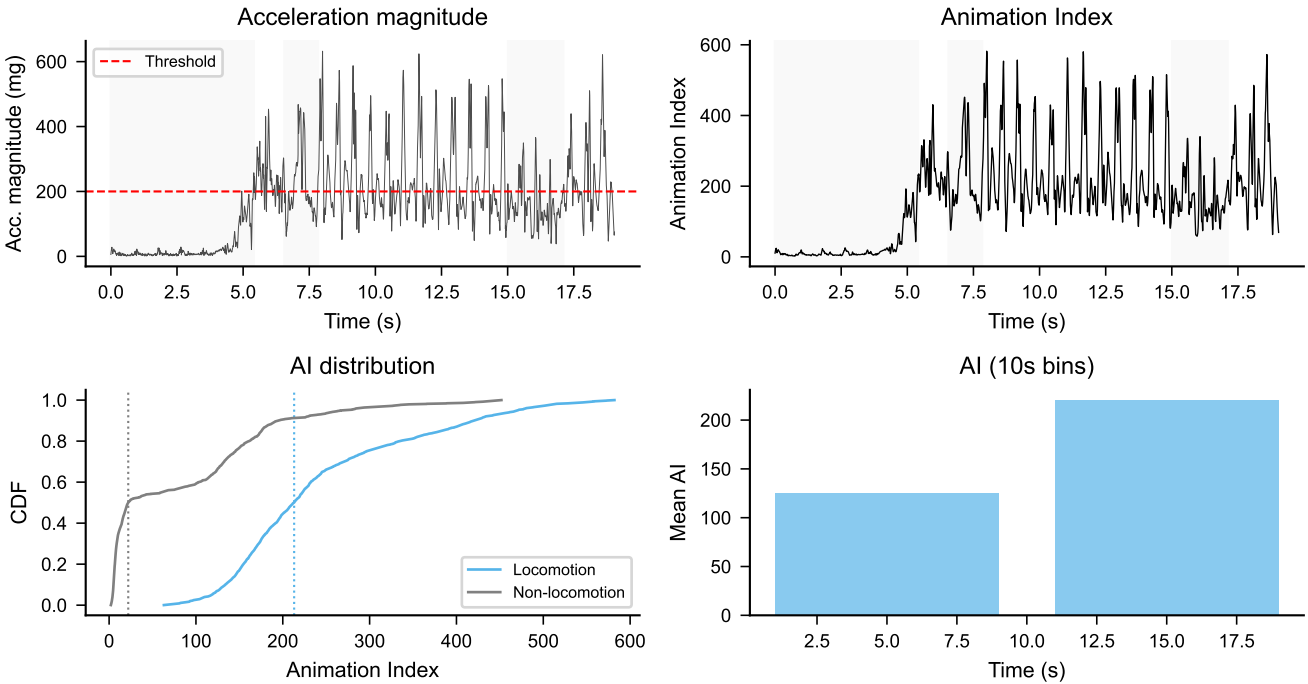
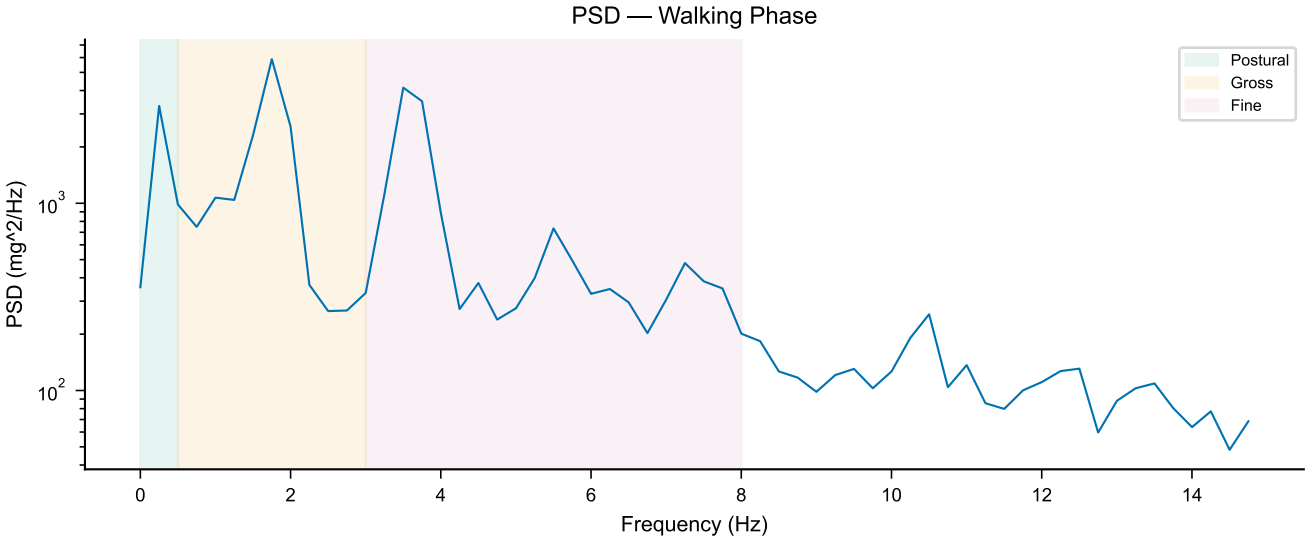
04 — Visualization Report

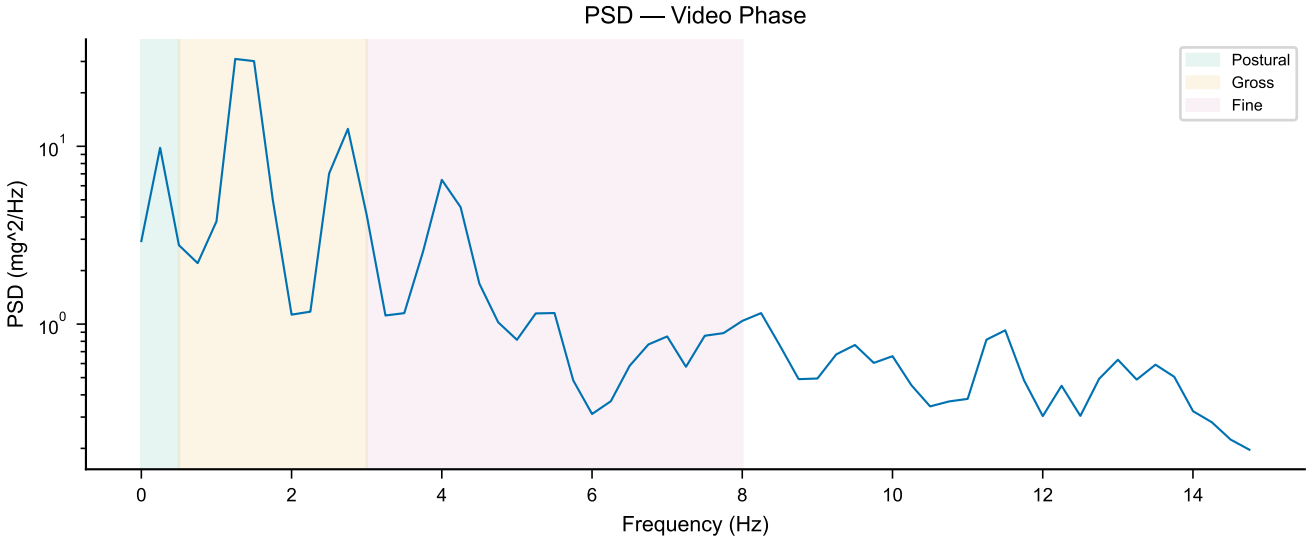
Publication-quality visualizations of movement-affect associations.

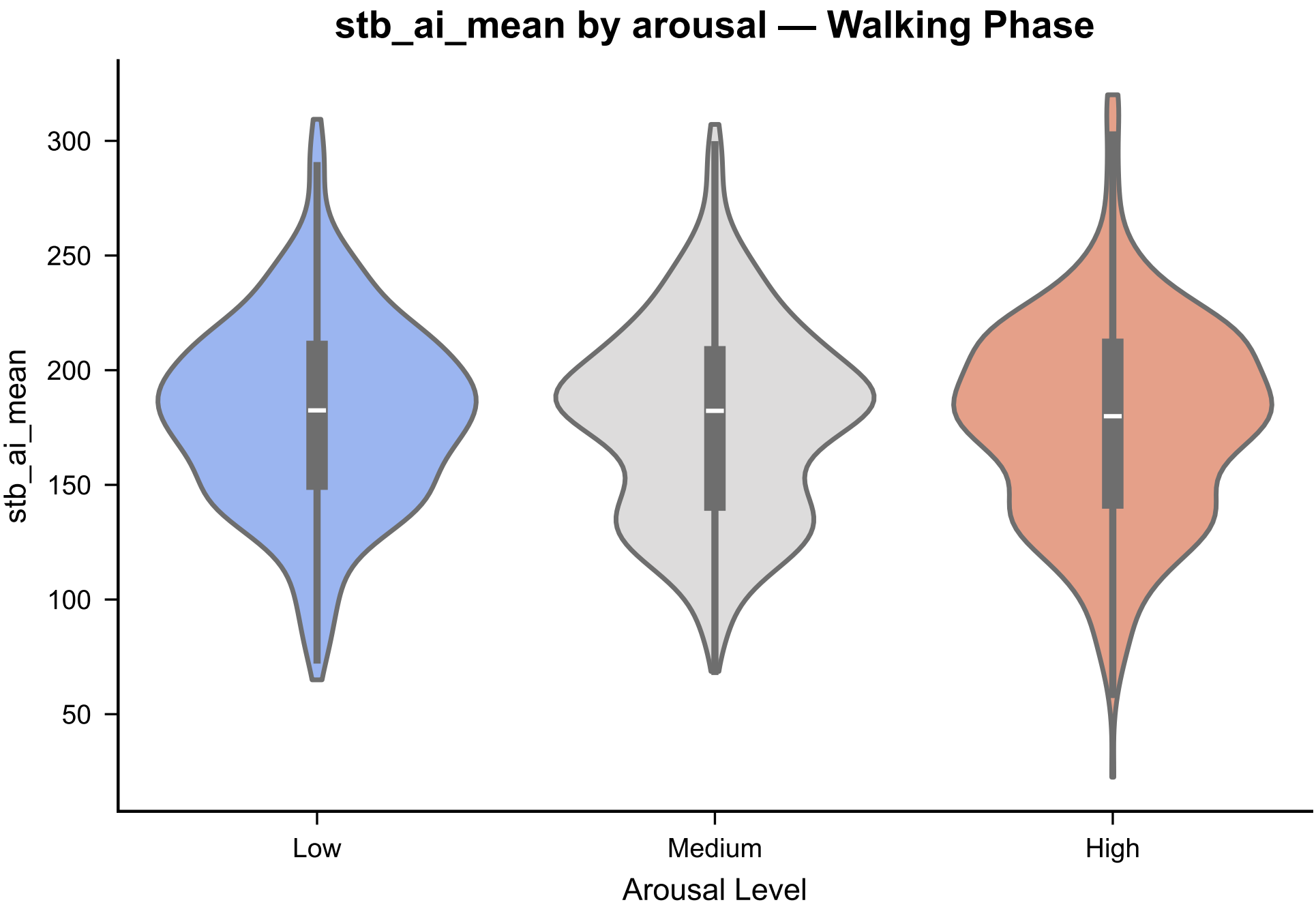
Contents:

1. Per-Trial Diagnostic Plots (representative participant)
2. Movement by Affect Level (violin plots)
3. Within-Participant Trends (spaghetti plots)
4. Phase Comparison Box Plots
5. Power Spectral Density Comparison

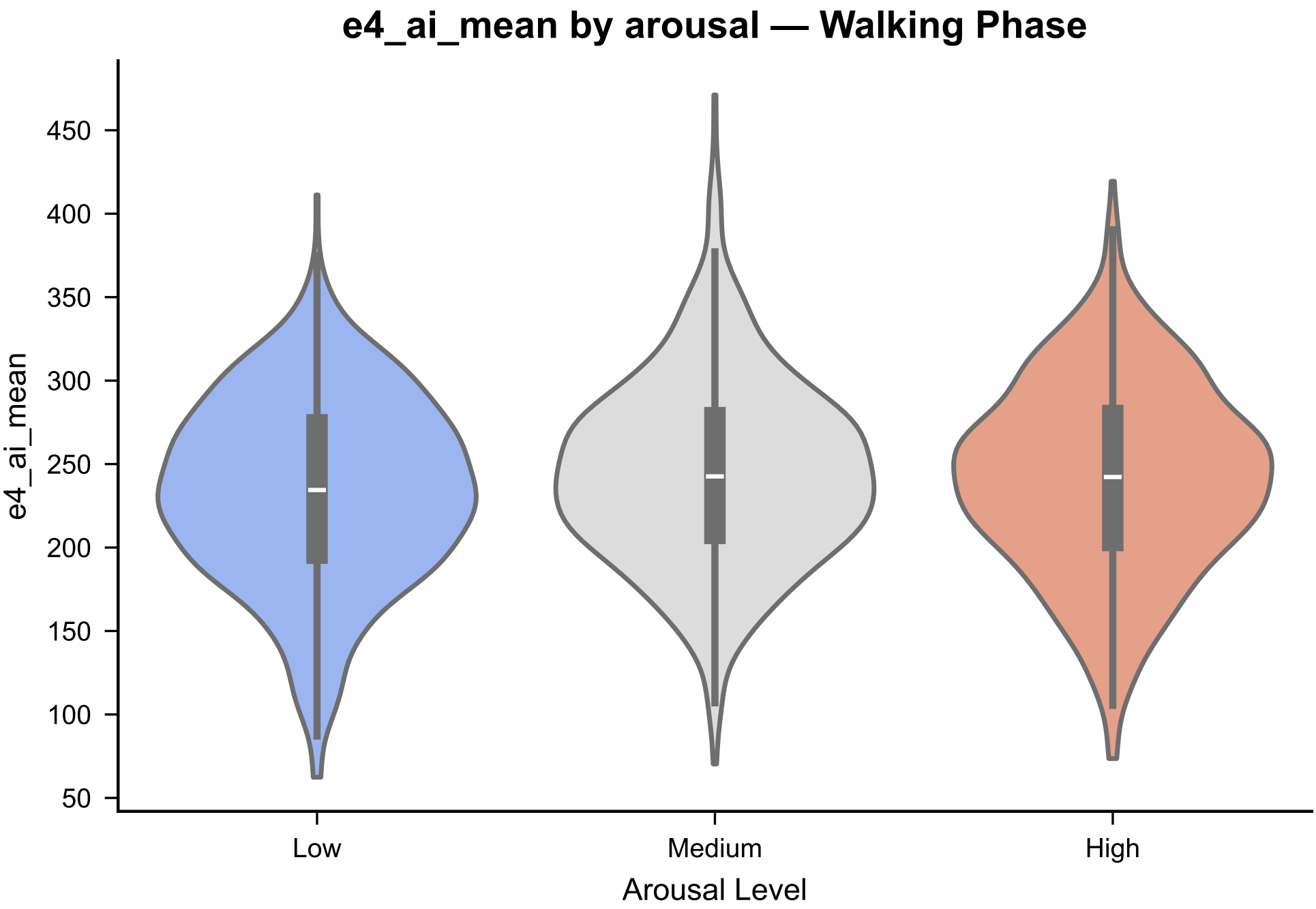

02-9TZO — Trial 1 — Walking
Acceleration magnitude
Threshold
Acc. magnitude (mg)
Time (s)
Animation Index
Animation Index
Time (s)
AI distribution
CDF
Locomotion
Non-locomotion
Animation Index
AI (10s bins)
Mean AI
Time (s)

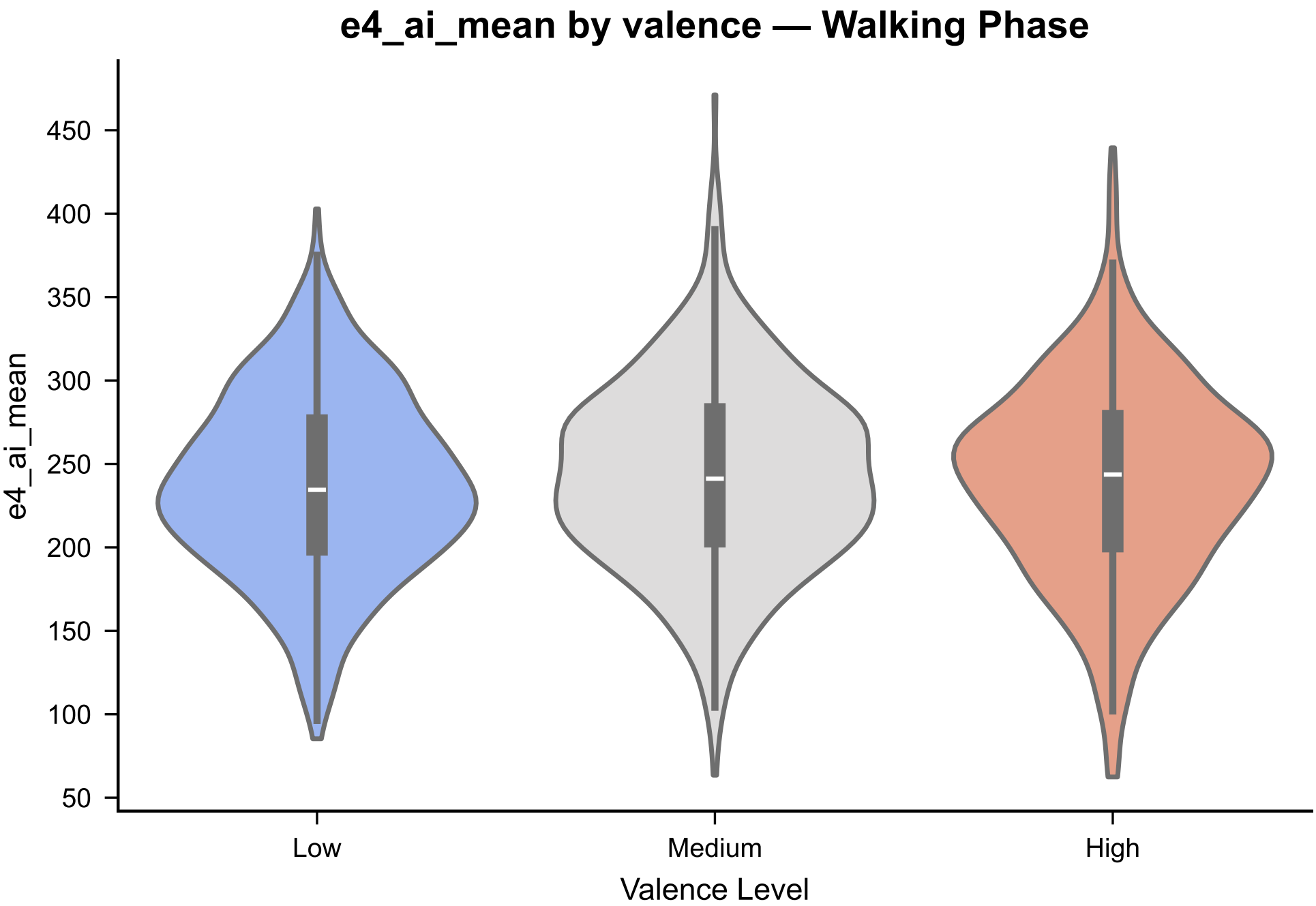

PSD — Walking Phase
Postural
Gross
Fine
PSD (mg^2/Hz)
10^3
10^2
0
2
4
6
8
10
12
14
Frequency (Hz)

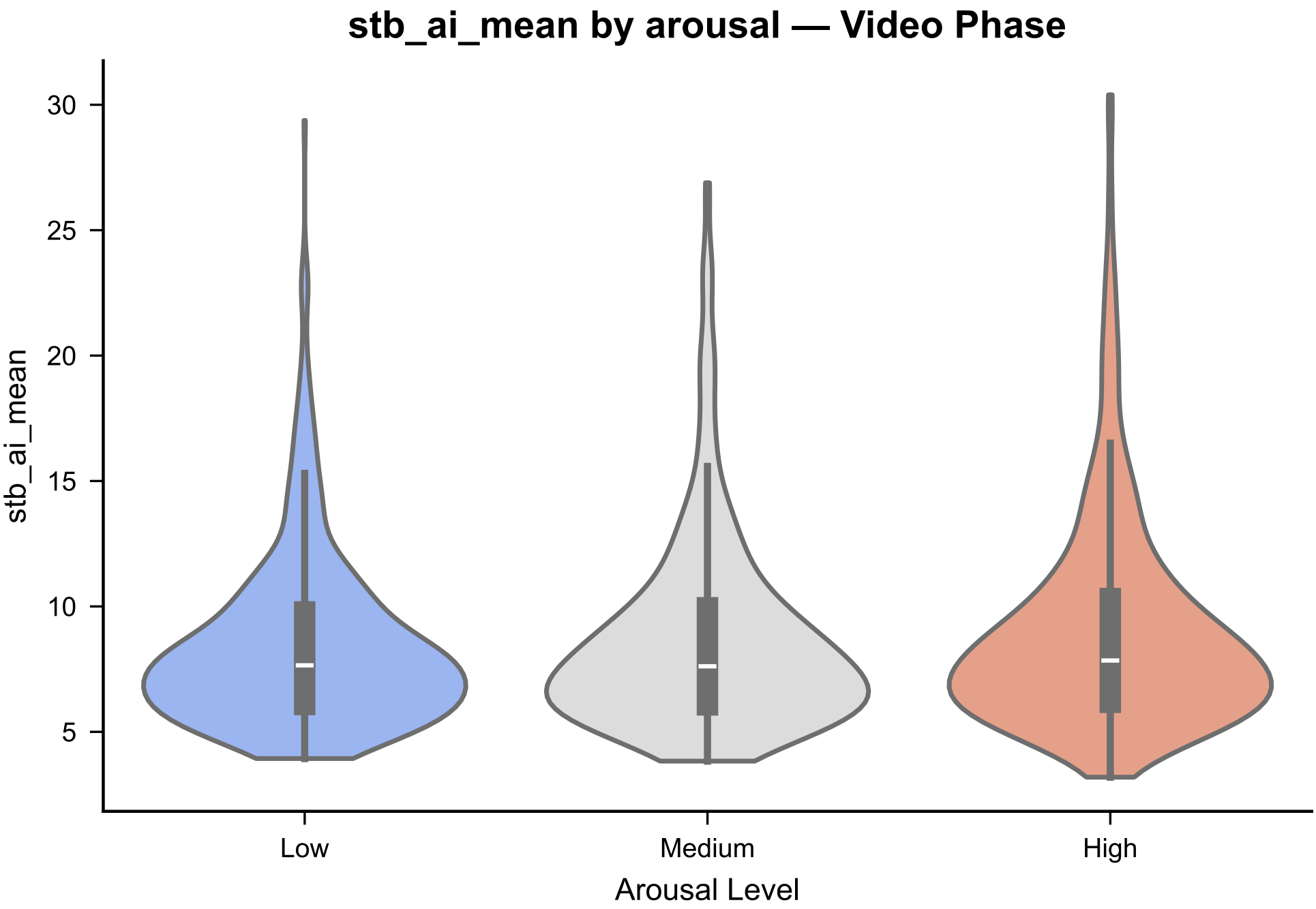

PSD — Video Phase
Postural
Gross
Fine
PSD (mg^2/Hz)
10^1
10^0
0
2
4
6
8
10
12
14
Frequency (Hz)

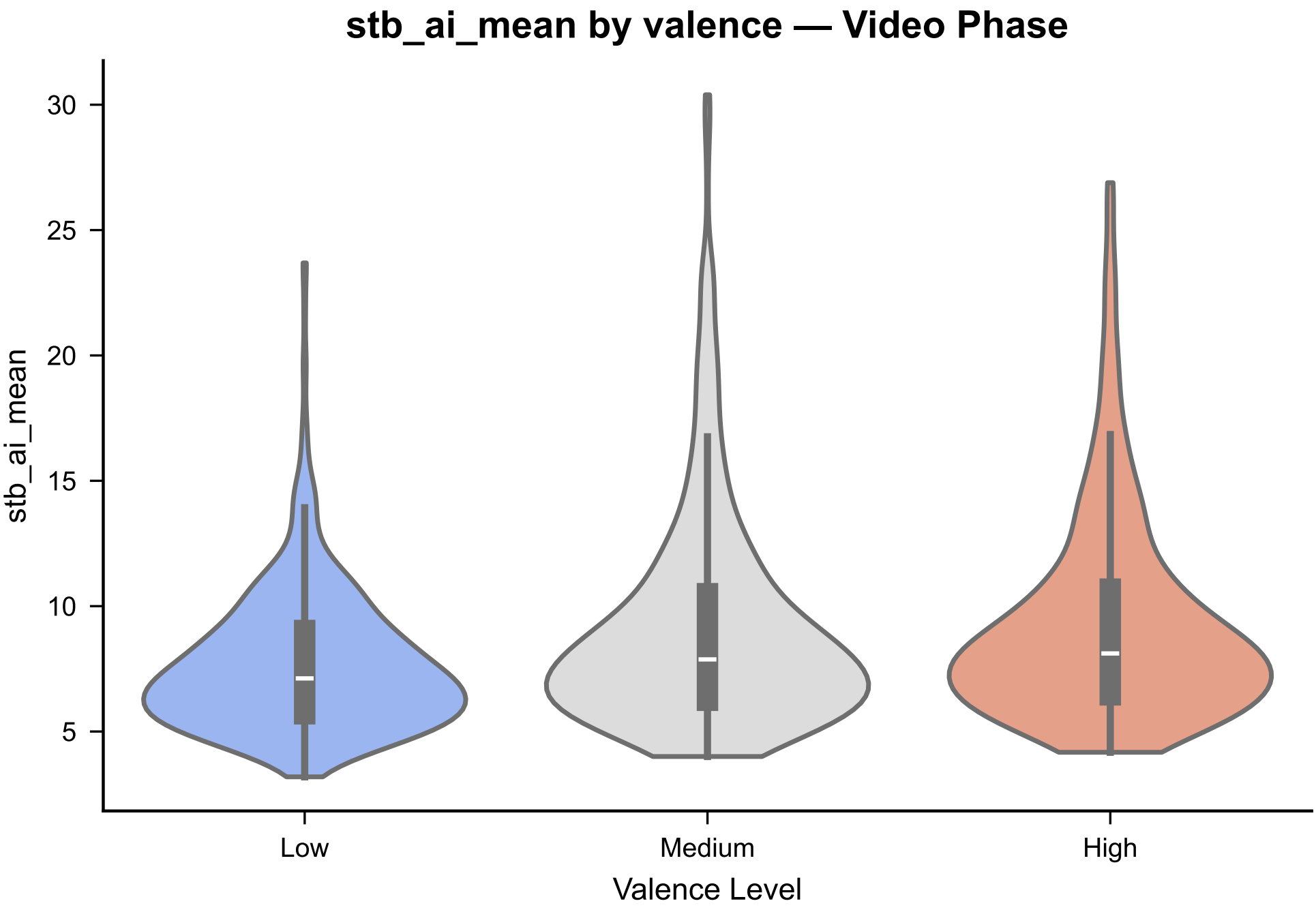

stb_ai_mean by arousal — Walking Phase
stb_ai_mean
300
250
200
150
100
50
Low
Medium
High
Arousal Level

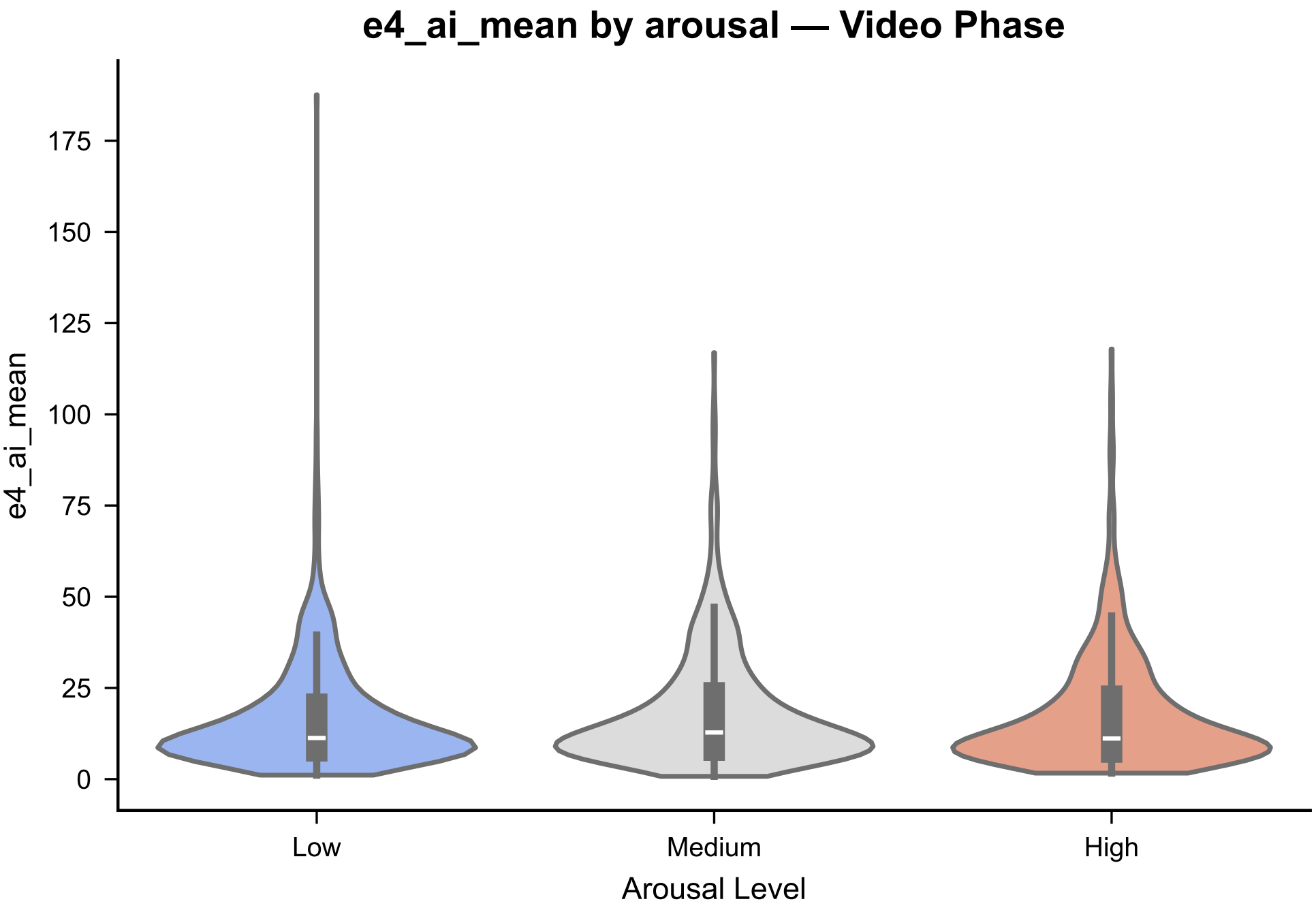
stb_ai_mean by valence — Walking Phase
stb_ai_mean
300
250
200
150
100
50
Low
Medium
High
Valence Level

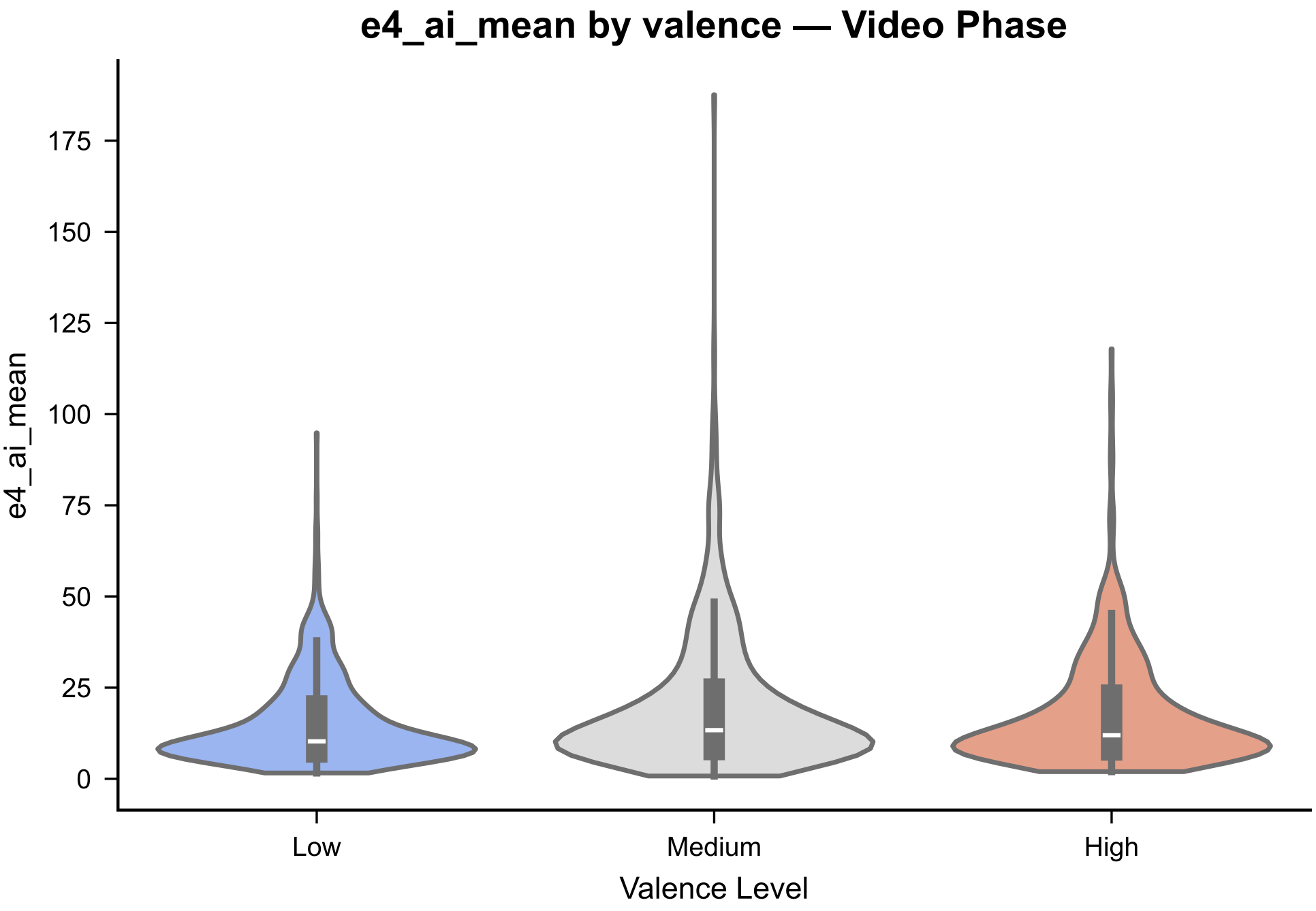

e4_ai_mean by arousal — Walking Phase
e4_ai_mean
450
400
350
300
250
200
150
100
50
Low
Medium
High
Arousal Level

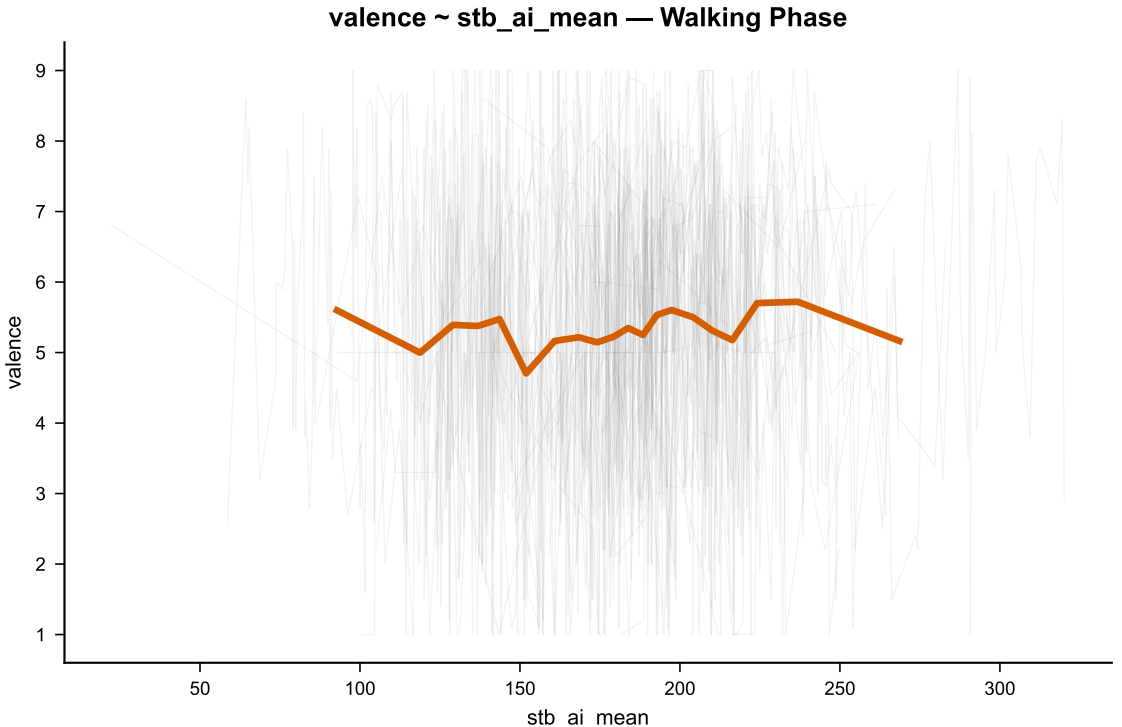

e4_ai_mean by valence — Walking Phase
e4_ai_mean
450
400
350
300
250
200
150
100
50
Low
Medium
High
Valence Level

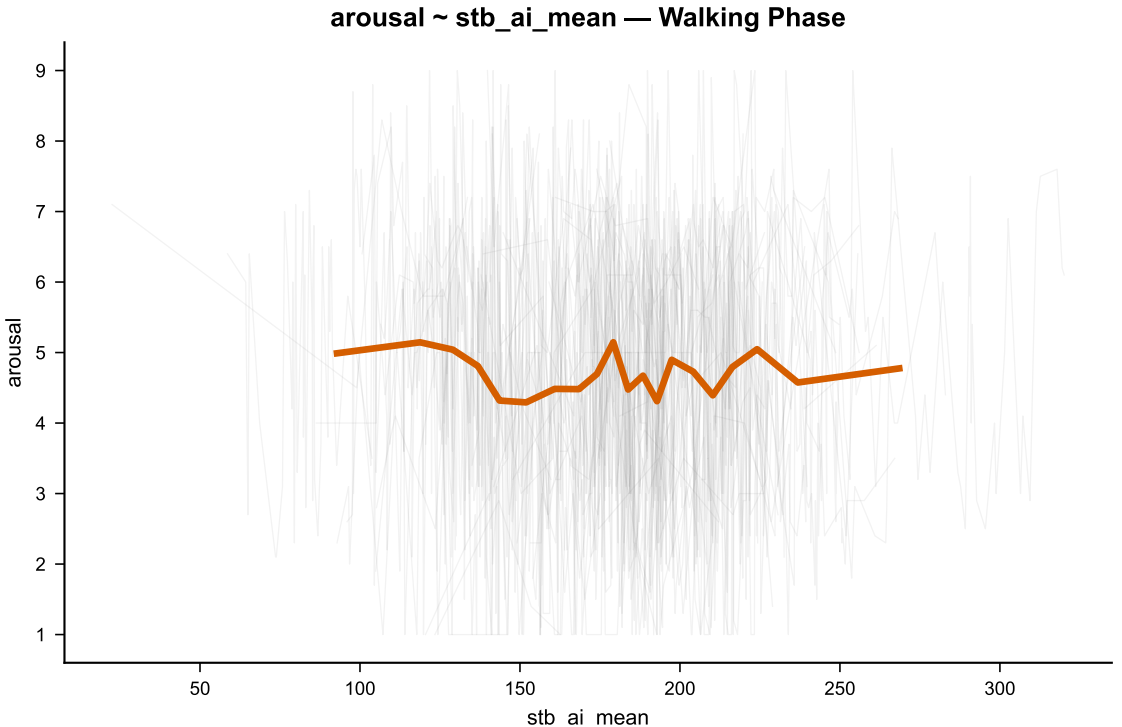

stb_ai_mean by arousal — Video Phase
stb_ai_mean
30
25
20
15
10
5
Low
Medium
High
Arousal Level

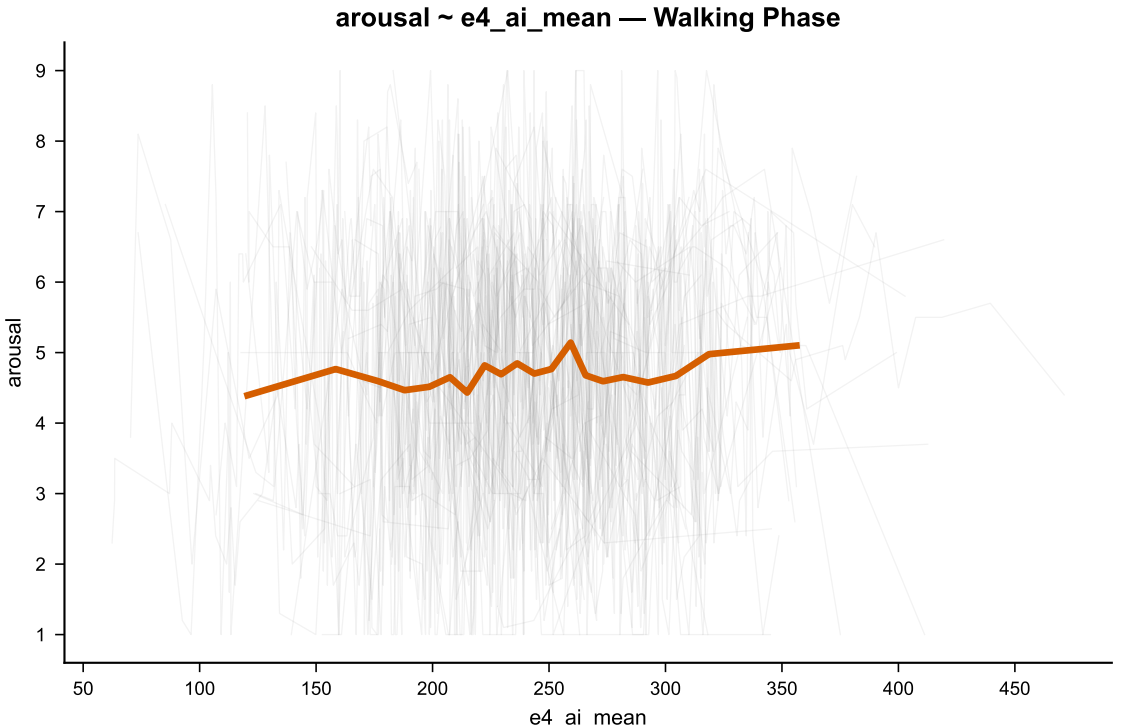

stb_ai_mean by valence — Video Phase
stb_ai_mean
30
25
20
15
10
5
Low
Medium
High
Valence Level

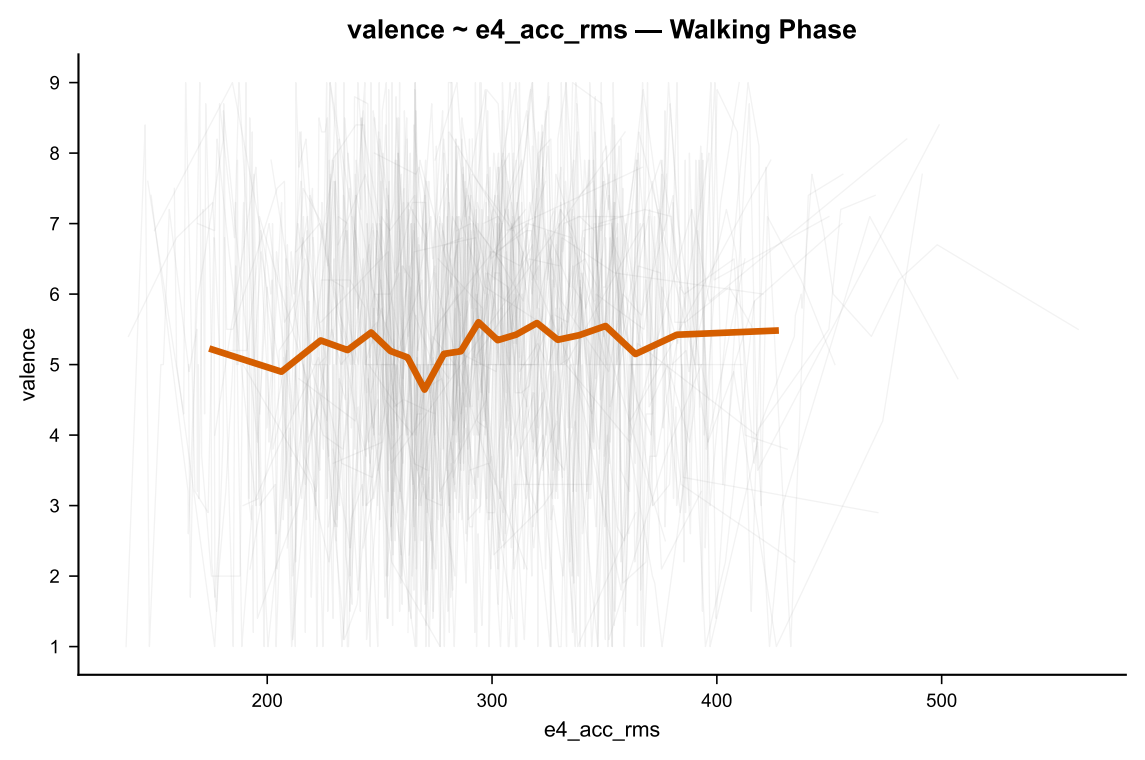

e4_ai_mean by arousal — Video Phase
e4_ai_mean
175
150
125
100
75
50
25
0
Low
Medium
High
Arousal Level

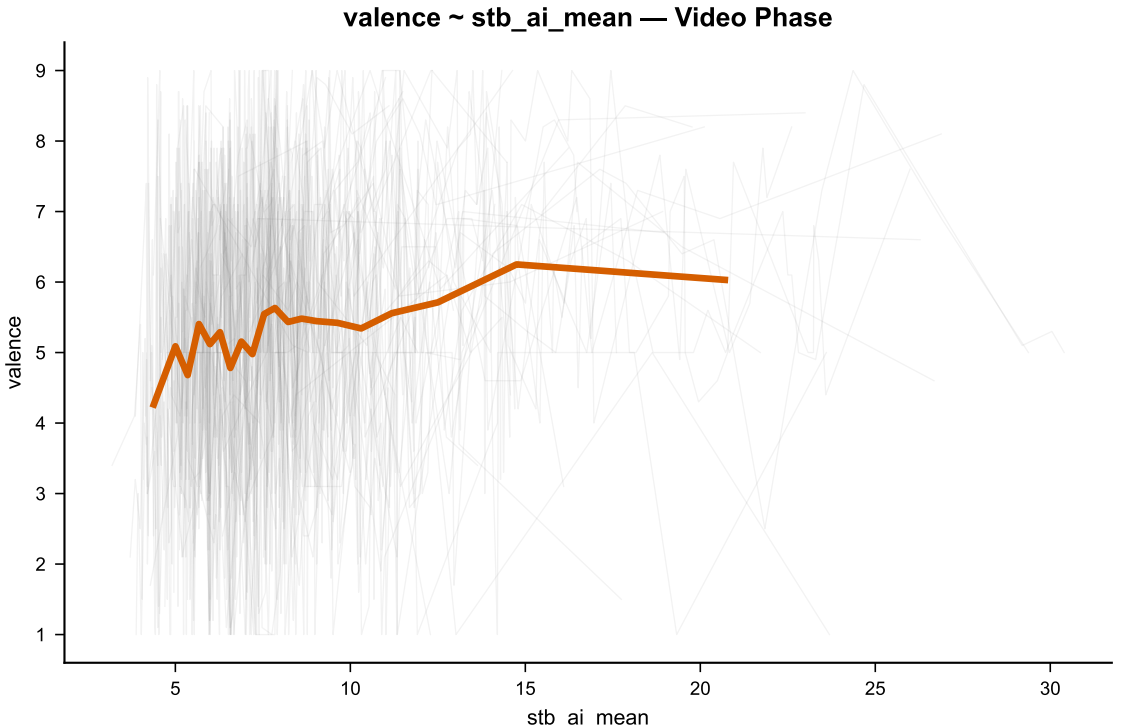

e4_ai_mean by valence — Video Phase
e4_ai_mean
175
150
125
100
75
50
25
0
Low
Medium
High
Valence Level

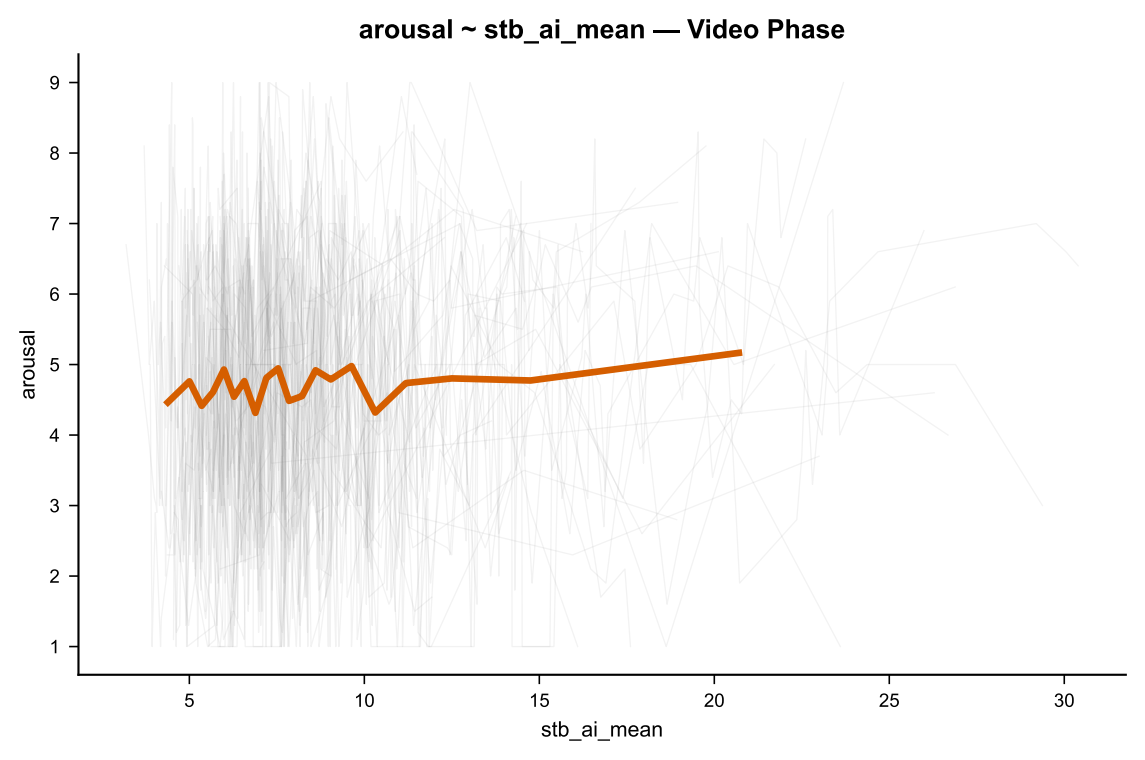

valence ~ stb_ai_mean — Walking Phase
valence
9
8
7
6
5
4
3
2
1
50
100
150
200
250
300
stb_ai_mean

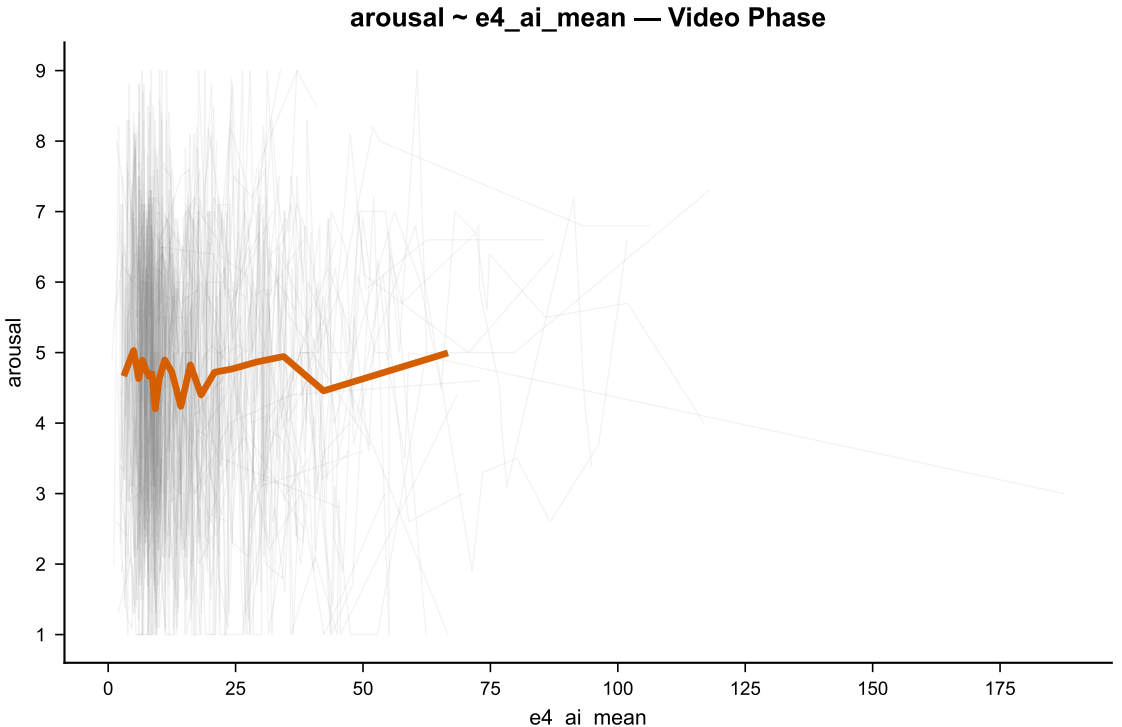

arousal ~ stb_ai_mean — Walking Phase
arousal
stb_ai_mean
1
2
3
4
5
6
7
8
9
50
100
150
200
250
300

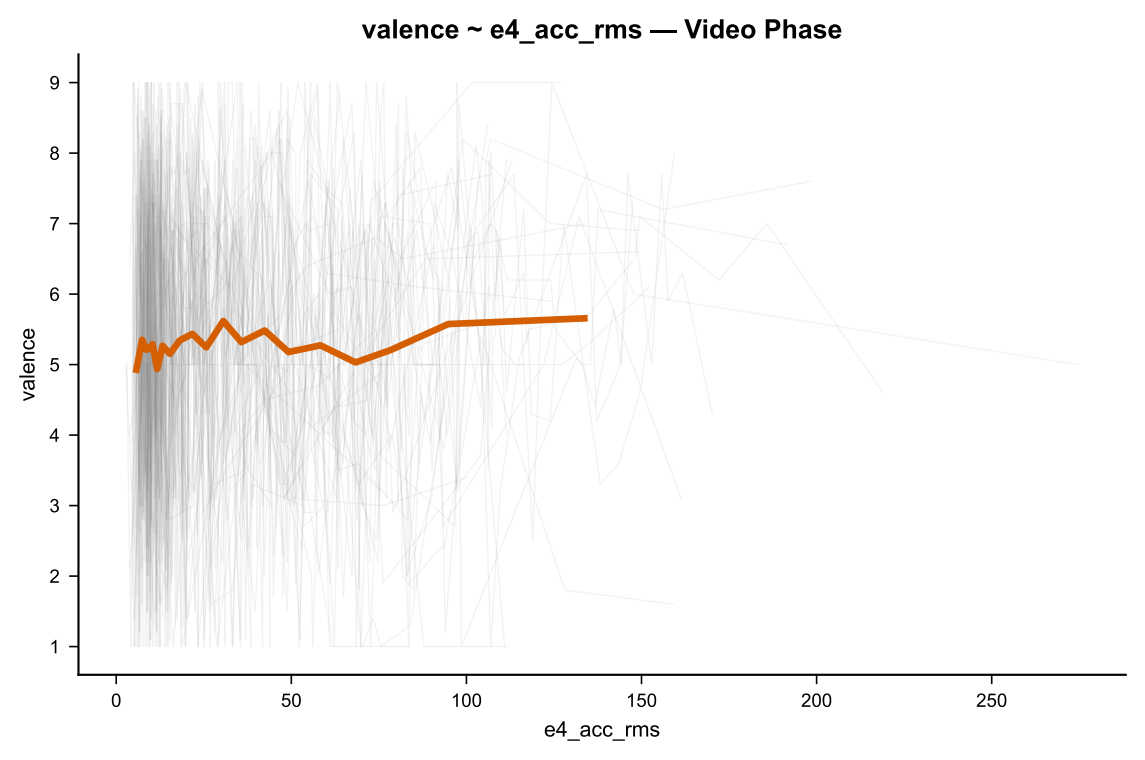

arousal ~ e4_ai_mean — Walking Phase
arousal
e4_ai_mean
1
2
3
4
5
6
7
8
9
50
100
150
200
250
300
350
400
450

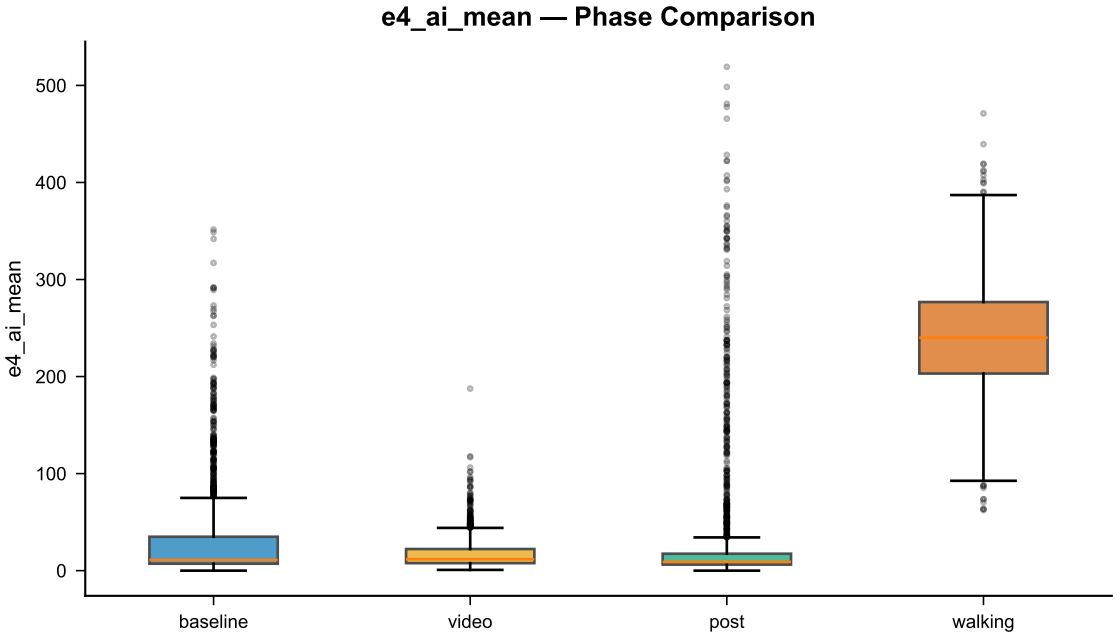

valence ~ e4_acc_rms — Walking Phase
valence
9
8
7
6
5
4
3
2
1
200
300
400
500
e4_acc_rms

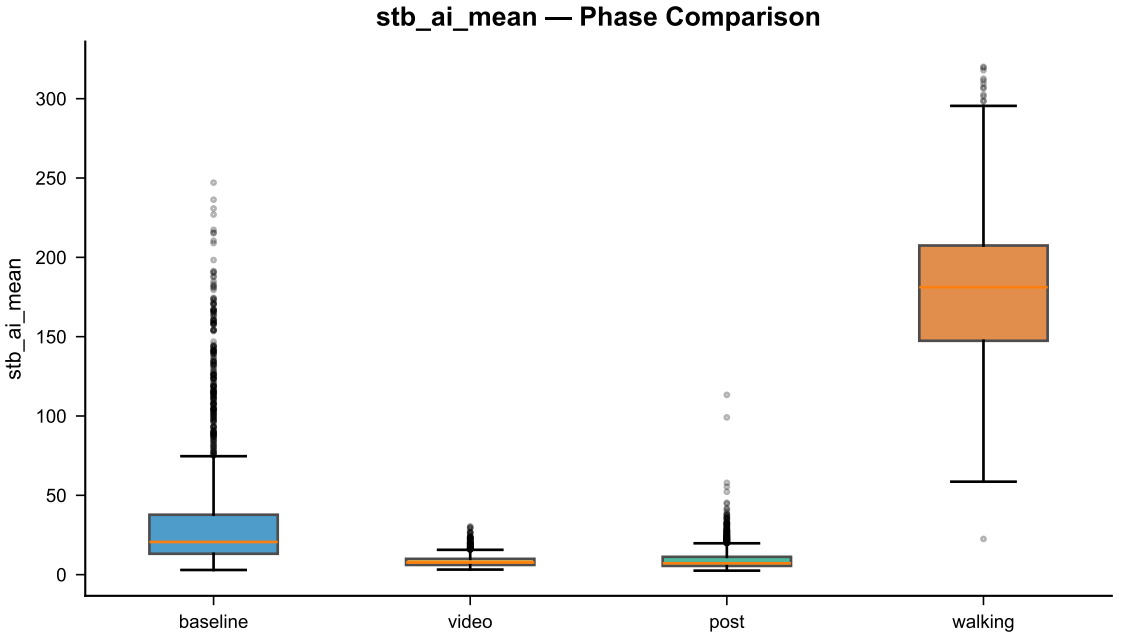

valence ~ stb_ai_mean — Video Phase
valence
stb_ai_mean
1
2
3
4
5
6
7
8
9
5
10
15
20
25
30

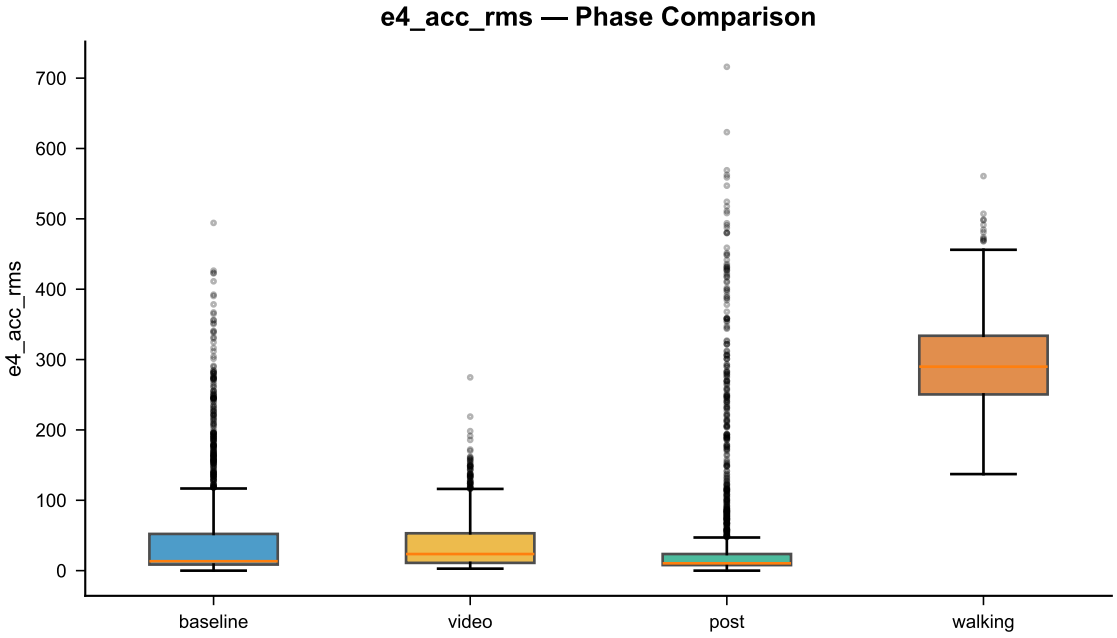

arousal ~ stb_ai_mean — Video Phase
arousal
stb_ai_mean
1
2
3
4
5
6
7
8
9
5
10
15
20
25
30

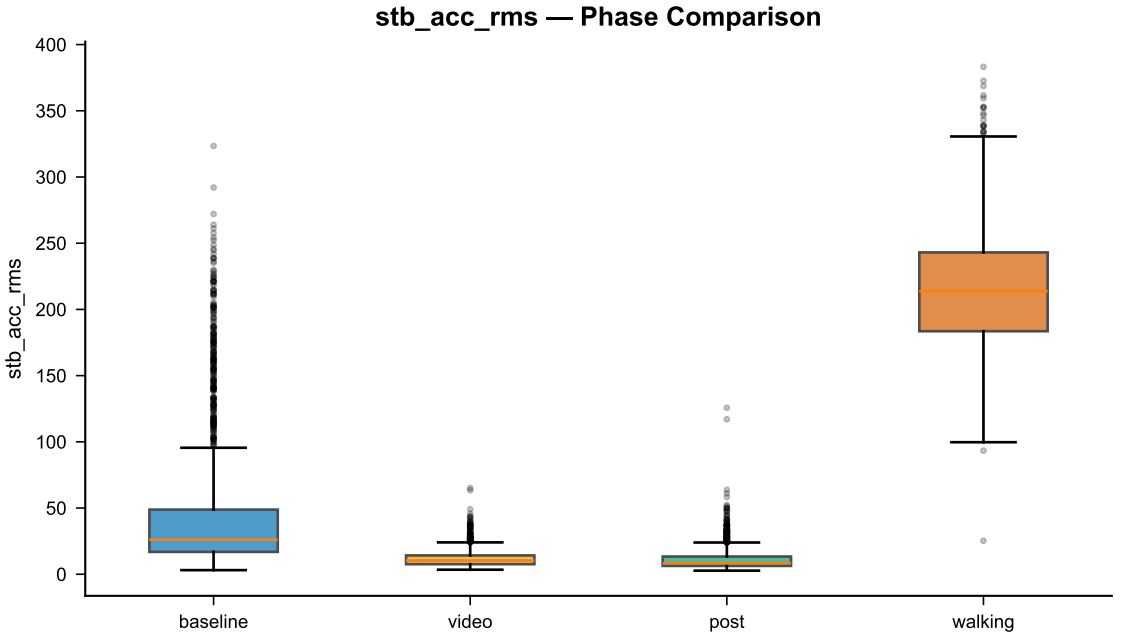

arousal ~ e4_ai_mean — Video Phase
arousal
9
8
7
6
5
4
3
2
1
0
25
50
75
100
125
150
175
e4_ai_mean


valence ~ e4_acc_rms — Video Phase
valence
e4_acc_rms
9
8
7
6
5
4
3
2
1
0
50
100
150
200
250


e4_ai_mean — Phase Comparison
e4_ai_mean
500
400
300
200
100
0
baseline
video
post
walking


stb_ai_mean — Phase Comparison
stb_ai_mean
300
250
200
150
100
50
0
baseline
video
post
walking


e4_acc_rms — Phase Comparison
e4_acc_rms
700
600
500
400
300
200
100
0
baseline
video
post
walking


stb_acc_rms — Phase Comparison
stb_acc_rms
400
350
300
250
200
150
100
50
0
baseline
video
post
walking